

CLINICAL TRIALS AND OBSERVATIONS

Daratumumab-bortezomib-cyclophosphamide-dexamethasone for newly diagnosed amyloidosis: ANDROMEDA final survival analysis

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KEY POINTS

- D-VCd resulted in deeper and more rapid hematologic responses and recovery of organ function, resulting in superior survival.
- The superior survival achieved with D-VCd supports and further solidifies this regimen as the standard of care for AL amyloidosis.

In the primary analysis of ANDROMEDA, addition of subcutaneous daratumumab to VCd (bortezomib, cyclophosphamide, and dexamethasone; D-VCd) significantly improved hematologic complete response (CR) rate vs VCd, establishing D-VCd as the only approved therapy for light-chain (AL) amyloidosis. We present results from the pre-planned final analysis. In this phase 3 trial, we randomly assigned 388 patients with newly diagnosed AL amyloidosis to 6 cycles of VCd alone (control group) or with subcutaneous daratumumab (D-VCd) followed by single-agent daratumumab every 4 weeks for up to 24 total cycles. The primary end point was hematologic CR. The updated hematologic CR rate was 59.5% for D-VCd vs 19.2% for VCd (odds ratio, 6.03; 95% confidence interval [CI], 3.80-9.58; $P < .0001$). Median time to hematologic CR was shorter with D-VCd (67.5 days [range, 8.0-879.0]) vs VCd (85.0 days [range, 14.0-617.0]). With a median follow-up of 61.4 months, significant improvement was observed with D-VCd vs VCd in major organ deterioration–progression-free survival (hazard ratio, 0.44; 95% CI,

0.31-0.63; $P < .0001$) and overall survival (hazard ratio, 0.62; 95% CI, 0.42-0.90; $P = .0121$). Cardiac and renal response rates were 2 to 3 times higher with D-VCd vs VCd. Achieving hematologic or cardiac CR was associated with improved major organ deterioration–progression-free survival and overall survival. Adverse events were consistent with the known safety profiles for VCd and daratumumab. Adding daratumumab to VCd resulted in deeper and more rapid hematologic responses and organ function recovery, translating to statistically significant improvement in both overall survival and major organ deterioration–progression-free survival in newly diagnosed AL amyloidosis. This trial was registered at www.clinicaltrials.gov as #NCT03201965.

Introduction

Systemic light-chain (AL) amyloidosis is characterized by the tissue deposition of immunoglobulin light chains, produced by clonal CD38⁺ plasma cells, as insoluble amyloid fibrils. Amyloid deposition in vital organs, primarily the heart and kidneys, leads to progressive organ dysfunction and can result in death.¹ In a longitudinal natural history study of patients with AL amyloidosis, the 5-year overall survival rate of those diagnosed from 2010 to 2019 was 48% and 35% for the group with cardiac involvement.² VCd (bortezomib, cyclophosphamide, and dexamethasone) was frequently used to treat newly diagnosed AL amyloidosis.³⁻⁵ Until recently, there was no approved treatment option available.

Daratumumab is a human immunoglobulin G κ monoclonal antibody targeting CD38 with direct on-tumor⁶⁻⁹ and immunomodulatory¹⁰⁻¹² mechanisms of action. The phase 3 ANDROMEDA study showed that the addition of subcutaneous daratumumab to VCd (D-VCd) significantly increased hematologic complete response (CR) rate vs VCd alone and prolonged major organ deterioration–progression-free survival in patients with newly diagnosed AL amyloidosis.¹³ These results led to the accelerated approval of D-VCd as the first and currently only regimen indicated for newly diagnosed AL amyloidosis.¹⁴⁻¹⁶

Here, we report results from the preplanned final analysis for major organ deterioration–progression-free survival and overall survival of ANDROMEDA.

Methods

Trial design and oversight

In this randomized, open-label, phase 3 study, patients were enrolled between 3 May 2018 and 15 August 2019 at 109 sites in 22 countries/territories across North and South America, Europe, the Middle East, and the Asia-Pacific region.

The study protocol was approved by an independent ethics committee or institutional review board at each site. The study was conducted in accordance with the International Council for Harmonisation Good Clinical Practice guidelines, the principles originating from the Declaration of Helsinki, and study site-specific regulations. All patients provided written informed consent.

Patients

Complete eligibility criteria have been previously published.¹³ Briefly, eligible patients were aged ≥ 18 years with a

histopathologic diagnosis of systemic AL amyloidosis (affecting ≥ 1 organ) and measurable hematologic disease. Patients were excluded if they had received previous therapy for AL amyloidosis, had symptomatic multiple myeloma according to International Myeloma Working Group criteria,¹⁷ had an Eastern Cooperative Oncology Group performance status score of >2 , had an estimated glomerular filtration rate of <20 mL/min per 1.73 m², or had evidence of a severe cardiovascular condition (including Mayo cardiac stage IIIB). Full eligibility criteria are provided in the study protocol.¹³

Trial treatments

Patients were randomly assigned (1:1) using an interactive web-response system to D-VCd or VCd. Randomization was stratified according to cardiac stage (I, II, or IIIA based on the European modification of the Mayo Clinic Cardiac Staging System³), availability of transplantation (countries that do or do not typically offer transplantation for AL amyloidosis), and renal function (creatinine clearance ≥ 60 or <60 mL/min). Treatment assignments were not blinded.

All patients were to receive subcutaneous bortezomib (1.3 mg/m²), oral or IV cyclophosphamide (300 mg/m²; 500 mg maximum weekly dose), and oral or IV dexamethasone (40 mg) once weekly for six 28-day cycles. Dexamethasone could be given at a dose of 20 mg weekly at the physician's discretion. Patients assigned to the D-VCd group also received subcutaneous daratumumab (1800 mg of daratumumab coformulated with recombinant human hyaluronidase PH20 [2000 U/mL; ENHANZE drug delivery technology, Halozyme, Inc, San Diego, CA]) weekly in cycles 1 and 2, every 2 weeks in cycles 3 through 6, and every 4 weeks thereafter until disease progression, the start of subsequent therapy, or a maximum of 24 cycles from the start of treatment, whichever occurred first. Details regarding preadministration and postadministration medications are provided in the supplemental Methods, available on the *Blood* website.

End points and assessments

The primary end point was overall hematologic CR (according to consensus guidelines¹⁸ as reported in supplemental Table 1), as assessed by the independent review committee. Secondary end points included major organ deterioration–progression-free survival (a composite end point including end-stage cardiac or renal failure, hematologic progression [assessed by the independent review committee], or death), organ response,^{19,20} overall survival, time to and duration of hematologic CR, and time to next treatment. The complete definitions of end points, disease evaluation timing, and

definitions of hematologic and organ response (including graded cardiac response criteria²¹) are provided in the supplemental Appendix (supplemental Methods and supplemental Tables 1-3).

Statistical analysis

The sample size was based on the assumption that the percentage of patients with hematologic CR would be 15 percentage points higher in the D-VCd group than in the VCd group; ~360 patients were required to provide 85% power to detect this difference (2-sided α level of 0.05). If the between-group difference for the primary end point was significant, the major secondary end points of major organ deterioration–progression-free survival and overall survival were sequentially tested using a hierarchical testing approach that controls the type I error.²²

Efficacy analyses were performed in the intention-to-treat population, which included all patients who underwent randomization. Organ response rates were assessed in the response-evaluable population. Response-evaluable patients at each time point who did not have an organ response assessment within the specified ± 1 -month window for any reason were considered nonresponders. The safety population included patients who received at least 1 dose of study treatment. Between-group differences in hematologic CR were tested using a stratified Cochran-Mantel-Haenszel test, and corresponding relative risk and odds ratios, 95% confidence intervals (CIs), and *P* values were reported. Time-to-event variables were evaluated using the Kaplan-Meier method. Major organ deterioration–progression-free survival was analyzed by employing an inverse-probability-of-censoring weighting method to adjust the estimates of a treatment effect in the presence of non-cross-resistant subsequent therapy (defined as any anti-plasma cell agent not included in the original protocol-assigned treatment).²³ The final analysis of major organ deterioration–progression-free survival, reported here, occurred when ~200 major organ deterioration–progression-free survival events had been observed. Full statistical methods are described in the supplemental Methods.

Results

Patients and treatment

A total of 388 patients were enrolled (D-VCd, *n* = 195; VCd, *n* = 193). Demographic and baseline characteristics were well balanced between groups and have been previously reported (supplemental Table 4).¹³ The median age was 64 years (range, 34-87) and median time since initial AL amyloidosis diagnosis to randomization was 43 days (range, 5-1611).

At the time of the primary analysis, 159 of 193 treated patients (82.4%) in the D-VCd group and 121 of 188 treated patients (64.4%) in the VCd group had received 6 cycles of treatment as specified in the protocol.¹³ At the current clinical cutoff (17 April 2024), all patients completed on-study treatment and 124 of 193 treated patients (64.2%) in the D-VCd group completed 2 years of daratumumab treatment (Figure 1). The most common reasons for treatment discontinuation in the D-VCd group were death (23 patients), receipt of subsequent therapy (17 patients), and adverse event (11 patients). The median duration

of study treatment was 21.3 months (range, 0.03-26.7) in the D-VCd group and 5.3 months (range, 0.03-7.3) in the VCd group.

Efficacy

Statistical significance for hematologic CR was established at the primary analysis,¹³ and updated hematologic CR rates are provided. At the current clinical cutoff, 116 patients (59.5%) in the D-VCd group and 37 patients (19.2%) in the VCd group achieved a hematologic CR (odds ratio, 6.03; 95% CI, 3.80-9.58; nominal *P* < .0001; Table 1). Among these 153 patients who achieved hematologic CR, 131 (99 in the D-VCd group and 32 in the VCd group) occurred within 6 months and 22 (17 in the D-VCd group and 5 in the VCd group) occurred after 6 months. Hematologic very good partial response or better rates were also higher for D-VCd vs VCd (79.0% vs 50.3%; nominal *P* < .0001; Table 1). Landmark analysis of hematologic CR rate at different time points in the intention-to-treat population showed higher hematologic CR rates with D-VCd vs VCd (50.3% vs 14.0% at 6 months, 47.7% vs 12.4% at 12 months, 47.2% vs 12.4% at 24 months, and 44.6% vs 9.3% at 36 months; all nominal *P* < .0001). Among responders, median time to hematologic CR was 67.5 days (range, 8.0-879.0) for D-VCd and 85.0 days (range, 14.0-617.0) for VCd; median time to first hematologic response (partial response or better) was 11.0 days (range, 5.0-145.0) for D-VCd and 23.0 days (range, 7.0-170.0) for VCd. The median duration of hematologic CR was not reached in either group. Median time to involved free AL (iFLC) level of ≤ 20 mg/L was shorter with D-VCd vs VCd (24.5 days [range, 5.0-372.0] vs 32.0 days [range, 8.0-150.0]), as was median time to a difference between the iFLC and uninvolved FLC (dFLC) levels of < 10 mg/L (30.0 days [range, 5.0-1114.0] vs 56.5 days [range, 8.0-1647.0]). Median iFLC and dFLC levels over time are shown in supplemental Figure 1.

A total of 235 patients (118 for D-VCd and 117 for VCd) were evaluable for cardiac response and 230 patients (117 for D-VCd and 113 for VCd) were evaluable for renal response. Cardiac response rates (Figure 2A) and renal response rates (Figure 2B) were higher with D-VCd vs VCd at 6, 12, 24, 36, and 48 months (all nominal *P* < .01). Among the 235 cardiac response–evaluable patients, 113 patients achieved cardiac very good partial response or better (corresponding to a $> 60\%$ reduction in NT-proBNP [N-terminal pro-B-type natriuretic peptide]; 76/118 patients [64.4%] for D-VCd and 37/117 patients [31.6%] for VCd); of these, 64 patients achieved cardiac CR (NT-proBNP reduction to ≤ 350 pg/mL; 48 [40.7%] for D-VCd and 16 [13.7%] for VCd). The median time to cardiac very good partial response or better was 14.9 months for D-VCd vs 16.4 months for VCd, and the median time to cardiac CR was 16.3 months for D-VCd vs 20.9 months for VCd.

At a median follow-up of 61.4 months (range, 0.0-71.2), a total of 197 major organ deterioration–progression-free survival events were observed (79 [40.5%] in the D-VCd group [41 hematologic progressions, 3 major organ deteriorations, and 35 deaths] and 118 [61.1%] in the VCd group [63 hematologic progressions, 11 major organ deteriorations, and 44 deaths]; Table 1). A total of 118 patients (31.0%; 33 [17.1%] in the D-VCd group and 85 [45.2%] in the VCd group) received non-cross-resistant subsequent therapy before or without experiencing hematologic progression (eg, patient may have had rising FLC levels but did not yet meet the criteria for

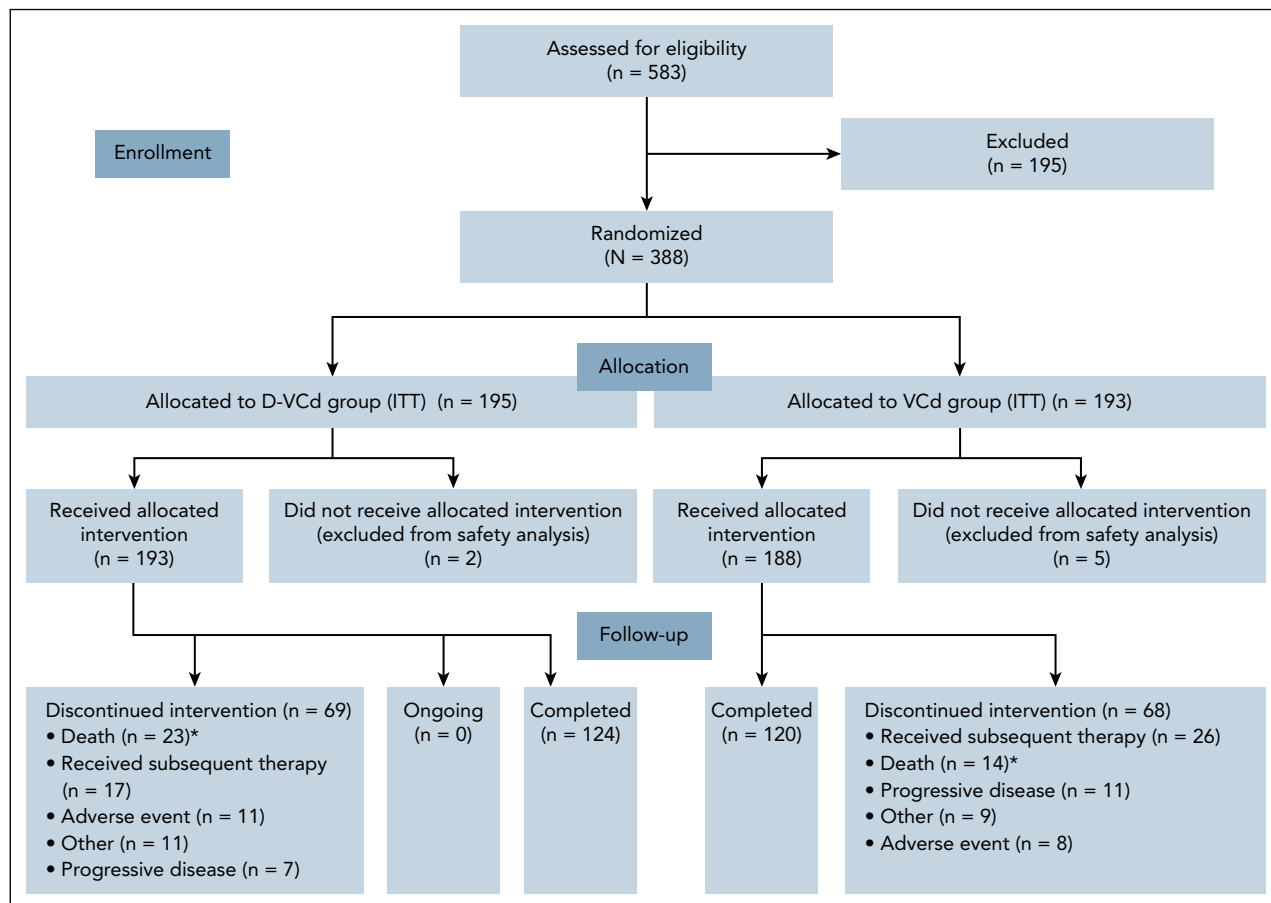


Figure 1. CONSORT patient flow diagram. *Deaths include only those that were the primary reason for treatment discontinuation, and do not represent the total number of deaths during the study period. ITT, intention to treat.

hematologic progression). After adjusting for dependent censoring due to switching to non-cross-resistant subsequent therapy, statistically significant improvement in major organ deterioration–progression-free survival was observed (hazard ratio, 0.44; 95% CI, 0.31-0.63; $P < .0001$ [crossing the pre-specified significance boundary of .0495]; Figure 3A). The hazard ratio for major organ deterioration–progression-free survival censoring non-cross-resistant subsequent therapy was 0.47 (95% CI, 0.33-0.67; $P < .0001$); the median was not reached for D-VCd vs 30.2 months (95% CI, 16.7-48.9) for VCd (Figure 3B). Other preplanned sensitivity/supplemental analyses of major organ deterioration–progression-free survival showed consistent results (supplemental Table 5). Subgroup analyses of major organ deterioration–progression-free survival demonstrated that the treatment effect of D-VCd over VCd was generally consistent across preplanned clinically relevant subgroups (supplemental Figure 2). Median hematologic progression-free survival was not reached for D-VCd vs 30.6 months (95% CI, 20.6-42.4) for VCd (hazard ratio, 0.49; 95% CI, 0.37-0.66; $P < .0001$; supplemental Figure 3).

Fifty of 193 patients (25.9%) in the D-VCd group and 115 of 188 patients (61.2%) in the VCd group received at least 1 non-cross-resistant subsequent therapy (supplemental Table 6). Of the 115 patients in the VCd arm who received non-cross-resistant subsequent therapy, 82 (71.3%) received subsequent

daratumumab-containing therapies. A total of 17 of 193 patients (8.8%) in the D-VCd group and 27 of 188 patients (14.4%) in the VCd arm received subsequent autologous stem cell transplantation.

A total of 112 deaths (46 [23.6%] in the D-VCd group and 66 [34.2%] in the VCd group) were observed. The hazard ratio for death was 0.62 (95% CI, 0.42-0.90; $P = .0121$ [crossing the prespecified stopping boundary of .0163]; Figure 4). The median overall survival was not reached in either group. Estimated 5-year overall survival rates were 76.1% (95% CI, 69.3-81.6) for D-VCd vs 64.7% (95% CI, 57.1-71.2) for VCd. The overall survival benefit of D-VCd over VCd was generally consistent across preplanned clinically relevant subgroups (supplemental Figure 4). Overall survival trended in favor of D-VCd, even when comparing with patients in the VCd arm who received daratumumab as subsequent therapy (hazard ratio, 0.82; 95% CI, 0.50-1.35).

Achieving hematologic CR was associated with improved major organ deterioration–progression-free survival and overall survival. At the 6-month landmark analysis, patients who achieved hematologic CR had better major organ deterioration–progression-free survival (hazard ratio, 0.30; 95% CI, 0.19-0.47) and overall survival (hazard ratio, 0.41; 95% CI, 0.23-0.72), regardless of treatment received (supplemental

Table 1. Summary of overall confirmed hematologic responses based on independent review committee assessment and major organ deterioration–progression-free survival and overall survival events in the intention-to-treat population

Variable	D-VCd (n = 195)	VCd (n = 193)	P value*
Hematologic response			
Any response, n	179	148	
Percent of patients (95% CI)	91.8 (87.0-95.2)	76.7 (70.1-82.5)	
CR, n†	116	37	<.0001
Percent of patients (95% CI)	59.5 (52.2-66.4)	19.2 (13.9-25.4)	—
Very good partial response or better, n (%)	154 (79.0)	97 (50.3)	—
Very good partial response, n (%)	38 (19.5)	60 (31.1)	—
Partial response, n (%)	25 (12.8)	51 (26.4)	—
No response, n (%)	8 (4.1)	38 (19.7)	—
Progressive disease, n (%)	0	0	—
Response could not be evaluated, n (%)	8 (4.1)	7 (3.6)	—
Major organ deterioration–progression-free survival, n (%)			
Hematologic progression	41 (21.0)	63 (32.6)	—
Major organ deterioration	3 (1.5)	11 (5.7)	—
Death	35 (17.9)	44 (22.8)	—
Overall survival, n (%)			
	46 (23.6)	66 (34.2)	—

Hematologic response was assessed centrally in the intention-to-treat population (defined as all patients who underwent randomization).

*P value was calculated with the use of the Cochran-Mantel-Haenszel χ^2 test.

†CR was based on consensus criteria with clarifications as specified in the trial protocol that required confirmation by the independent review committee. CR was defined as negative immunofixation and normalization of the FLC ratio without confirmation,¹⁹ a reduction in the absolute iFLC level to ≤ 20 mg/L,²⁴ and a reduction in the dFLC to < 10 mg/L.⁵

Figure 5). Similarly, patients who achieved cardiac CR had better major organ deterioration–progression-free survival (hazard ratio, 0.23; 95% CI, 0.12-0.43) and overall survival (hazard ratio, 0.05; 95% CI, 0.01-0.19), regardless of treatment received (supplemental Figure 6). In a combined analysis, patients who achieved hematologic CR or cardiac CR had better major organ deterioration–progression-free survival than patients who did not achieve hematologic CR and cardiac CR (hazard ratio, 0.17; 95% CI, 0.11-0.25). Major organ deterioration–progression-free survival was also evaluated by hematologic CR and cardiac CR, with both comparisons analyzed by treatment arm. Achieving hematologic or cardiac CR was associated with improved major organ deterioration–progression-free survival across treatment arms (Figure 5).

Safety

The safety profile of D-VCd was consistent with that observed in the primary analysis.¹³ The most common adverse events of any grade (occurring in $>25\%$ of patients in either group) and of grade 3 or 4 (occurring in $\geq 5\%$ of patients in either group) are summarized in Table 2. The most common grade 3 or 4 adverse events were lymphopenia (13.0% for D-VCd and 10.1% for VCd), pneumonia (8.3% and 4.3%), diarrhea (5.7% and 3.7%), cardiac failure (6.2% and 2.7%), neutropenia (5.2% and 2.7%), syncope (6.2% and 6.4%), fatigue (5.2% and 3.2%), hypokalemia (2.1% and 5.3%), and peripheral edema (3.1% and 5.9%). Twenty-six (13.5%) patients in the D-VCd group and 18 (9.6%) patients in the VCd group had grade ≥ 2 peripheral sensory neuropathy.

Serious adverse events occurred in 47.2% of patients in the D-VCd group and 36.2% of patients in the VCd group (supplemental Table 7), the most common of which were pneumonia (7.3% for D-VCd and 4.8% for VCd) and cardiac failure (7.3% and 4.3%, respectively). The rate of treatment discontinuation due to adverse events was 5.2% for D-VCd and 4.3% for VCd; the most common adverse events leading to treatment discontinuation were cardiac failure (0.5% for D-VCd and 1.1% for VCd) and fatigue (0.5% and 1.1%, respectively). Death due to adverse events within 30 days of the last study treatment occurred in 22 patients (11.4%) in the D-VCd group and 14 patients (7.4%) in the VCd group. Most of these deaths due to adverse events (16/22 patients in the D-VCd group and 12/14 patients in the VCd group) were unrelated to study treatment, and most deaths due to adverse events overall in both treatment arms occurred in patients with cardiac involvement at baseline. Systemic administration-related reactions with daratumumab occurred in 14 patients (7.3%); all were grade 1 or 2 and most (86%) occurred during the first injection.

Discussion

With a median follow-up of 61.4 months, representing 50 additional months of survival follow-up after the primary analysis of ANDROMEDA,¹³ significant and clinically meaningful improvement in major organ deterioration–progression-free survival and overall survival were observed with D-VCd vs VCd. To our knowledge, this is the first study comparing 2 contemporary regimens that has shown a survival improvement for

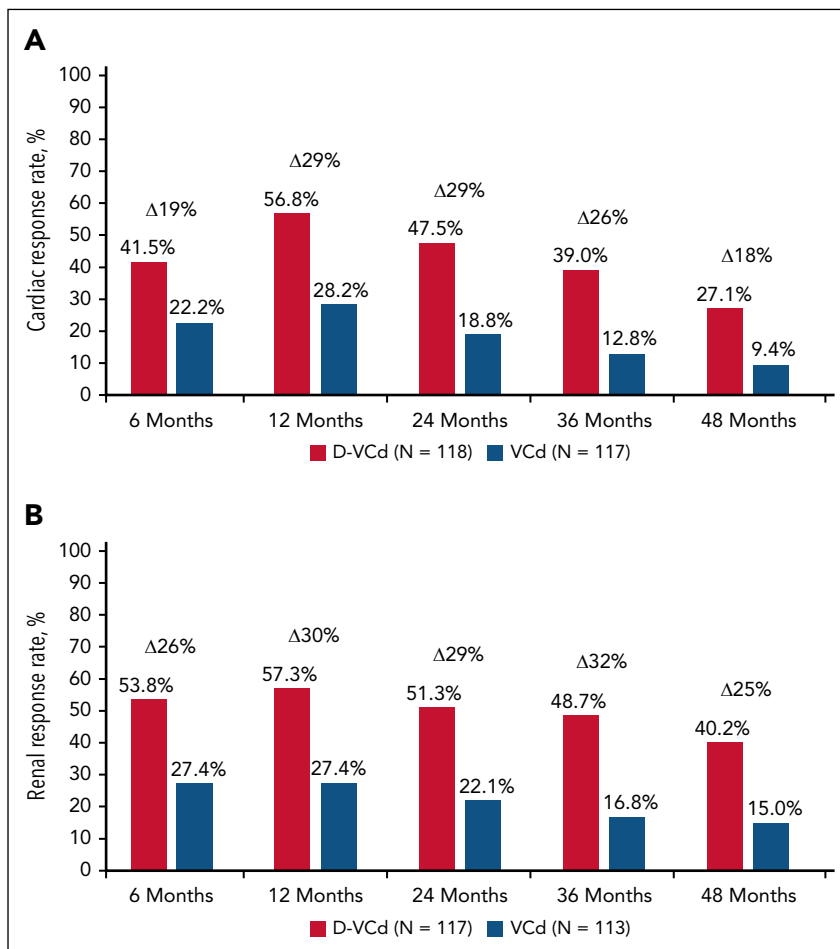


Figure 2. Organ response rates. (A) Cardiac response rates by the independent review committee assessment. (B) Renal response rates by the independent review committee assessment. Cardiac and renal response rates at a specific time point were calculated as the number of patients who had cardiac/renal response at the specific time point within a 1-month window; the denominator remained unchanged at each time point and represents the response-evaluable population. The cardiac/renal response rates displayed here are results without censoring non-cross-resistant anti-plasma cell therapy.

patients with AL amyloidosis. Subgroup analyses of major organ deterioration–progression-free survival and overall survival were generally consistent across preplanned clinically relevant subgroups, including patients with stage IIIA cardiac disease and those with t(11;14) translocations. However, the results observed in some subgroups should be interpreted with caution due to the small sample sizes and low number of events.

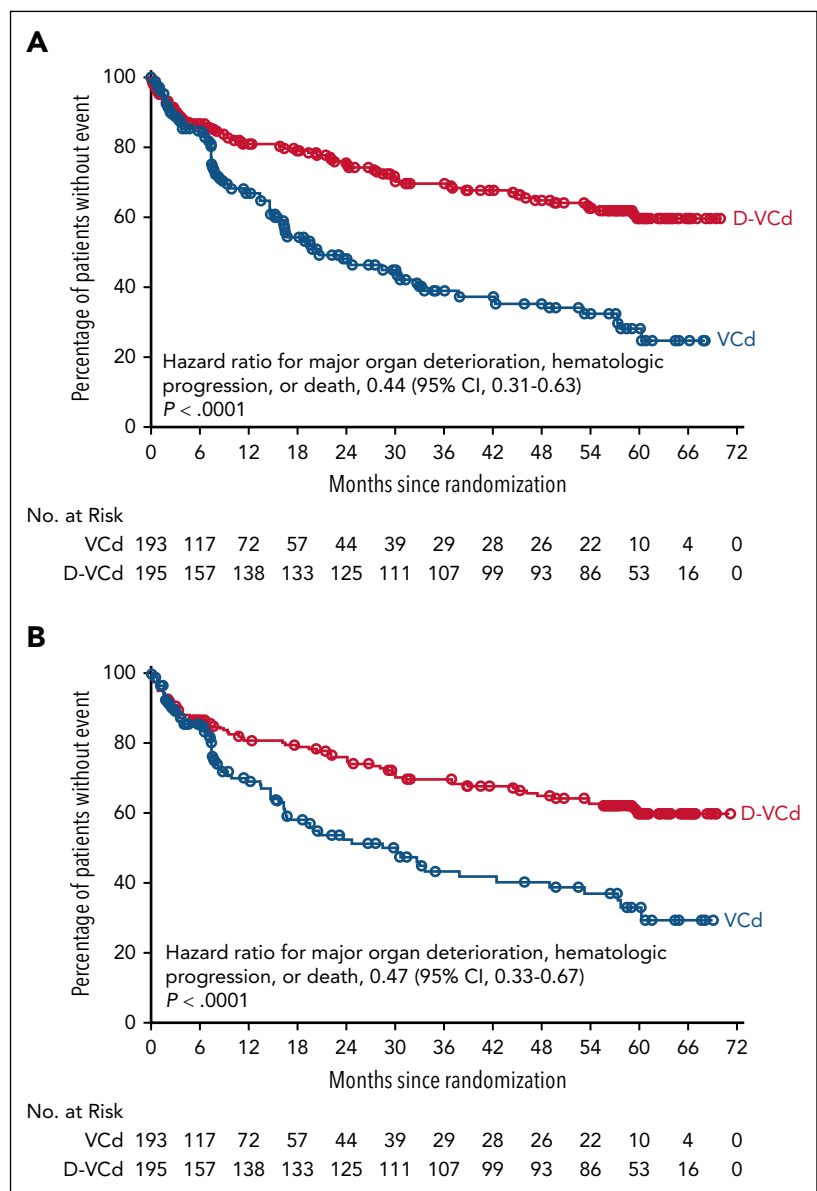
The treatment effect of D-VCd over VCd for the primary end point of overall hematologic CR continued to improve at the major organ deterioration–progression-free survival final analysis. Median time to hematologic CR was ~2.5 weeks shorter with D-VCd than with VCd (67.5 vs 85.0 days). This is of particular value in patients with advanced cardiac involvement in whom rapid and deep hematologic response translates to significantly improved survival.²⁵ The proportion of patients who achieved hematologic CR was >3.5 times higher with D-VCd at 6, 12, and 24 months and was almost 5 times higher at 36 months vs VCd. Of the 153 patients who achieved hematologic CR (116 for D-VCd vs 37 for VCd), most occurred within 6 months (99 for D-VCd vs 32 for VCd), with an additional 22 patients (17 for D-VCd vs 5 for VCd) achieving hematologic CR after 6 months. Moreover, overall survival trended in favor of D-VCd, even when comparing with patients in the VCd arm who received daratumumab as subsequent therapy. These observations reinforce the widely accepted principle that using the most effective regimen up front leads to long-term clinical benefits, including improvements in major

organ deterioration–progression-free survival and overall survival. The currently ongoing phase 2 EMILIA study ([ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT05898646) identifier: NCT05898646) is evaluating 6 cycles vs 18 cycles of daratumumab maintenance in patients with AL amyloidosis who have responded to 6 cycles of D-VCd induction therapy.

Cardiac and renal response rates at 6, 12, 24, 36, and 48 months were 2 to 3 times higher for patients in the D-VCd arm than for those in the VCd arm. It is important to note that the denominator for all time points during which organ response rates were assessed was the response-evaluable population, which remained unchanged at each time point. However, patients without an organ response assessment within the specified ± 1 -month window for any reason were considered nonresponders. In addition, cardiac CR rate was nearly tripled with the addition of daratumumab to VCd vs VCd alone. This is particularly noteworthy because patients who achieve a complete cardiac response regain the life expectancy of the matched general population, which is a definition of “operational cure.”²⁶

Importantly, achieving hematologic CR or cardiac CR was associated with improved major organ deterioration–progression-free survival and overall survival, irrespective of the treatment received, confirming the earlier established strong correlation between the depth of hematologic response and long-term outcomes,²⁷ such as overall survival, and the

Figure 3. Major organ deterioration–progression-free survival. (A) Results of the Kaplan-Meier estimates of major organ deterioration–progression-free survival based on the independent review committee assessment and inverse-probability-of-censoring weighting method among patients in the intention-to-treat population. (B) Results of the Kaplan-Meier estimates of major organ deterioration–progression-free survival censoring non-cross-resistant subsequent therapy. Major organ deterioration was defined as end-stage cardiac or renal failure.



clinical relevance of hematologic CR as an end point for the evaluation of new therapies for AL amyloidosis. The treatment effect of major organ deterioration–progression-free survival was also evaluated by hematologic CR or cardiac CR and separated by treatment arm. The greatest benefits in major organ deterioration–progression-free survival were observed in patients treated with D-VCd who achieved hematologic CR or cardiac CR, which were more frequently achieved in the D-VCd arm than in the VCd arm.

Due to the study design, which included the continuation of daratumumab in the D-VCd group beyond the fixed 6 cycles of VCd in both groups, the median treatment duration was ~16 months longer in the D-VCd group (21.3 months vs 5.3 months for VCd) and represented an additional 11.7 months of treatment for the D-VCd group compared with the primary analysis.¹³ No new safety signals were identified with extended follow-up. The safety profile of D-VCd was consistent with that observed in the primary analysis and remains consistent with the known safety profiles of VCd and daratumumab. Although

some adverse events occurred more frequently in the D-VCd group (eg, any-grade peripheral sensory neuropathy, any-grade upper respiratory tract infection, and grade 3 or 4 pneumonia and cardiac failure), the incidence of adverse events should be interpreted in the context of the substantially longer treatment exposure for patients in the D-VCd group.

In conclusion, the addition of daratumumab to VCd resulted in deeper and more rapid hematologic responses and the recovery of organ function with a higher cardiac CR rate, resulting in a clinically meaningful and statistically significant improvement in both overall survival and major organ deterioration–progression-free survival. The safety profile of D-VCd was manageable and consistent with the known safety profile of daratumumab and VCd. The initial results of ANDROMEDA changed the treatment paradigm in patients with AL amyloidosis, and the results of the final analysis demonstrate the superior survival achieved with D-VCd, supporting and further solidifying the benefits of this regimen as the standard of care in this difficult-to-treat disease.

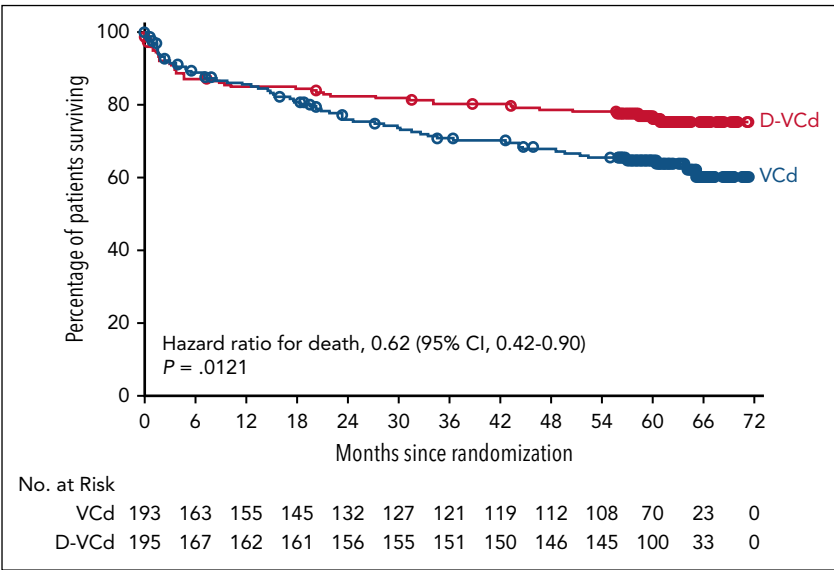


Figure 4. Overall survival. Results of the Kaplan-Meier estimates of overall survival among patients in the intention-to-treat population.

Figure 5. Correlation between HemCR or CarCR and major organ deterioration–progression-free survival by treatment arm. (A) Kaplan-Meier estimates of major organ deterioration–progression-free survival for patients who achieved HemCR vs those who did not achieve HemCR by treatment arm at a 6-month landmark analysis. (B) Kaplan-Meier estimates of major organ deterioration–progression-free survival, censored for non-cross-resistant subsequent therapy, for patients who achieved CarCR vs those who did not achieve CarCR by treatment arm. There were 8 patients who achieved CarCR after receiving non-cross-resistant subsequent therapy; these 8 patients were treated as non-CarCR for the evaluation of major organ deterioration–progression-free survival. Major organ deterioration was defined as end-stage cardiac or renal failure. CarCR, cardiac complete response; HemCR, hematologic complete response.

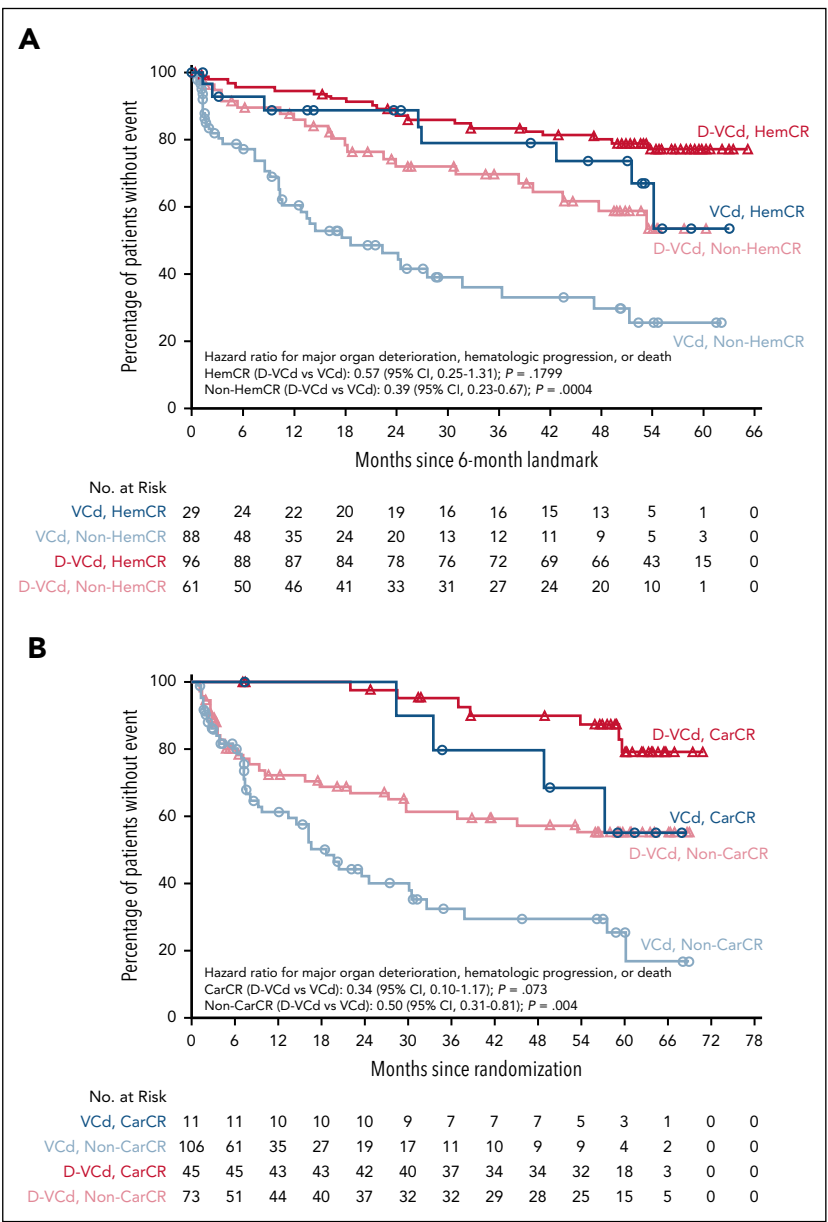


Table 2. Most common adverse events during treatment in the safety population

Event	D-VCd (n = 193)		VCd (n = 188)	
	Any grade, n (%)	Grade 3 or 4, n (%)	Any grade, n (%)	Grade 3 or 4, n (%)
Peripheral edema	71 (36.8)	6 (3.1)	68 (36.2)	11 (5.9)
Diarrhea	70 (36.3)	11 (5.7)	57 (30.3)	7 (3.7)
Constipation	70 (36.3)	3 (1.6)	54 (28.7)	0
Peripheral sensory neuropathy	65 (33.7)	5 (2.6)	37 (19.7)	4 (2.1)
Fatigue	55 (28.5)	10 (5.2)	53 (28.2)	6 (3.2)
Nausea	55 (28.5)	3 (1.6)	52 (27.7)	0
Upper respiratory tract infection	50 (25.9)	1 (0.5)	21 (11.2)	1 (0.5)
Anemia	49 (25.4)	8 (4.1)	44 (23.4)	9 (4.8)
Insomnia	49 (25.4)	0	47 (25.0)	2 (1.1)
Dyspnea	49 (25.4)	5 (2.6)	32 (17.0)	6 (3.2)
Lymphopenia	37 (19.2)	25 (13.0)	28 (14.9)	19 (10.1)
Hypokalemia	26 (13.5)	4 (2.1)	28 (14.9)	10 (5.3)
Pneumonia	24 (12.4)	16 (8.3)	12 (6.4)	8 (4.3)
Neutropenia	21 (10.9)	10 (5.2)	12 (6.4)	5 (2.7)
Cardiac failure	18 (9.3)	12 (6.2)	10 (5.3)	5 (2.7)
Syncope	16 (8.3)	12 (6.2)	12 (6.4)	12 (6.4)

The safety population included patients who received at least 1 dose of study treatment. Adverse events of any grade that were reported in >25% of patients in either treatment group and grade 3 or 4 adverse events that were reported in at least 5% of patients in either treatment group are listed.

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Authorship

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The online version of this article contains a data supplement.

There is a *Blood Commentary* on this article in this issue.

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